

RailSuperApp

Unified Railway Travel, Booking & Wallet Platform

Project Report

Aim

To design and develop a unified railway travel super-app, **RailSuperApp**, that integrates ticket booking (reserved, unreserved, and platform tickets), real-time train tracking, in-app wallet and payment management, on-board food ordering, complaint/support handling, and an AI-powered chatbot assistant into a single platform — providing passengers with a seamless, end-to-end railway travel experience.

Objectives

- 1 To develop a secure user authentication system with registration, login, and biometric verification.
- 2 To implement a train search and booking module supporting reserved, unreserved, and platform tickets across multiple trains per route, including daily and weekly-frequency trains.
- 3 To design an in-app wallet system enabling top-ups, secure payments, and accurate tracking of spent and credited amounts.
- 4 To provide real-time train tracking, showing live train location, delay estimates, and coach position on the platform.
- 5 To integrate an on-board food ordering system allowing passengers to order meals delivered directly to their seat.
- 6 To build a complaint and support module for passengers to raise and track service-related issues.
- 7 To develop an AI-powered chatbot capable of handling common queries and escalating complex issues to human support.
- 8 To ensure the platform is scalable and reliable under concurrent user load during peak booking periods.
- 9 To provide a multilingual, user-friendly interface accessible across devices.

Features

User Management

- Registration & login with biometric/SSO authentication
- Multilingual user interface
- User profile & document management

Ticket Booking

- Reserved, unreserved, and platform ticket booking
- Multiple trains per route with daily and weekly running schedules
- PNR status lookup
- Ticket cancellation with refund processing

Wallet & Payments

- In-app wallet with balance top-up
- Secure payment gateway integration
- Accurate transaction ledger (debit/credit tracking)
- Real-time spend and balance updates

Live Train Tracking

- Real-time train location tracking
- Delay estimates and ETA updates
- Coach position finder on the platform

Food Ordering

- Vendor menu browsing by station

- Seat-linked food ordering
- Order status tracking

Support & Complaints

- Complaint registration linked to bookings
- Complaint status tracking
- AI chatbot for instant query resolution
- Escalation to human support when needed

Dashboard

- Overview of bookings, confirmed trips, wallet balance, and total spend
- Quick access to all modules

Workflow

- 1 Register / Login — JWT authentication
- 2 Search trains — by route and date
- 3 Select train, book seat — class and passenger details
- 4 Pay via wallet — debit logged, balance updated
- 5 Booking confirmed — PNR generated
- 6 During the journey — track train, order food, and access support/chatbot in parallel

Technology Stack & Versions

Core Runtime

Component	Version
Java	17+
Maven	3.9+
Node.js	18+
MySQL	8
Spring Boot	3.3

Backend Dependencies (pom.xml)

Dependency	Purpose
spring-boot-starter-web	REST APIs
spring-boot-starter-data-jpa	ORM / database access
spring-boot-starter-security	Auth (BCrypt, JWT)
spring-boot-starter-websocket	Live tracking updates
spring-boot-starter-validation	Request validation
mysql-connector-j	MySQL JDBC driver
jjwt	JWT token generation/validation
lombok	Boilerplate reduction

Frontend

Tool	Notes
React (Vite)	Frontend framework/build tool
npm	Package manager
Zustand or Context	Auth state management
axios	API service calls

Pages

#	Page	Purpose
1	LoginPage	User login with JWT auth
2	RegisterPage	New user registration, form validation
3	DashboardPage	User summary cards, quick access links
4	SearchTrainsPage	Search trains by from/to/date, view availability
5	BookingPage	Seat class selection, passenger details, wallet payment
6	MyBookingsPage	List of bookings with status, cancel action, food orders
7	PnrStatusPage	Look up booking status by PNR number
8	WalletPage	Balance display, top-up modal, transaction history
9	TransactionsPage	Full transaction ledger
10	TrackingPage	Real-time train location via WebSocket
11	FoodOrderPage	Browse menu by train, add to cart, checkout
12	SupportPage	Create/view support tickets

API Endpoints

Identity / Auth

Method	Endpoint	Purpose
POST	/api/auth/register	Register new user
POST	/api/auth/login	Login, returns JWT

Ticketing

Method	Endpoint	Purpose
GET	/api/trains/search	Search trains by route/date
POST	/api/bookings	Create a booking
GET	/api/bookings/pnr/{pnr}	Get booking status by PNR
POST	/api/bookings/{id}/cancel	Cancel a booking

Payments / Wallet

Method	Endpoint	Purpose
GET	/api/wallet/balance	Get current wallet balance
POST	/api/wallet/topup	Add funds to wallet
GET	/api/wallet/transactions	Get transaction ledger
POST	/api/checkout	Deduct wallet balance, confirm booking

Tracking

Method	Endpoint	Purpose
GET	/api/tracking/{trainId}	Get current train position
WS	tracking channel	Live position push updates

Food

Method	Endpoint	Purpose
GET	/api/food/menu/{trainId}	Get food menu for a train

POST	/api/food/order	Place a food order
------	-----------------	--------------------

Support / Chatbot

Method	Endpoint	Purpose
POST	/api/support/tickets	Create a support ticket
GET	/api/support/tickets	List support tickets
POST	/api/chatbot/message	Send a message, get bot reply

UML Diagram of Entities

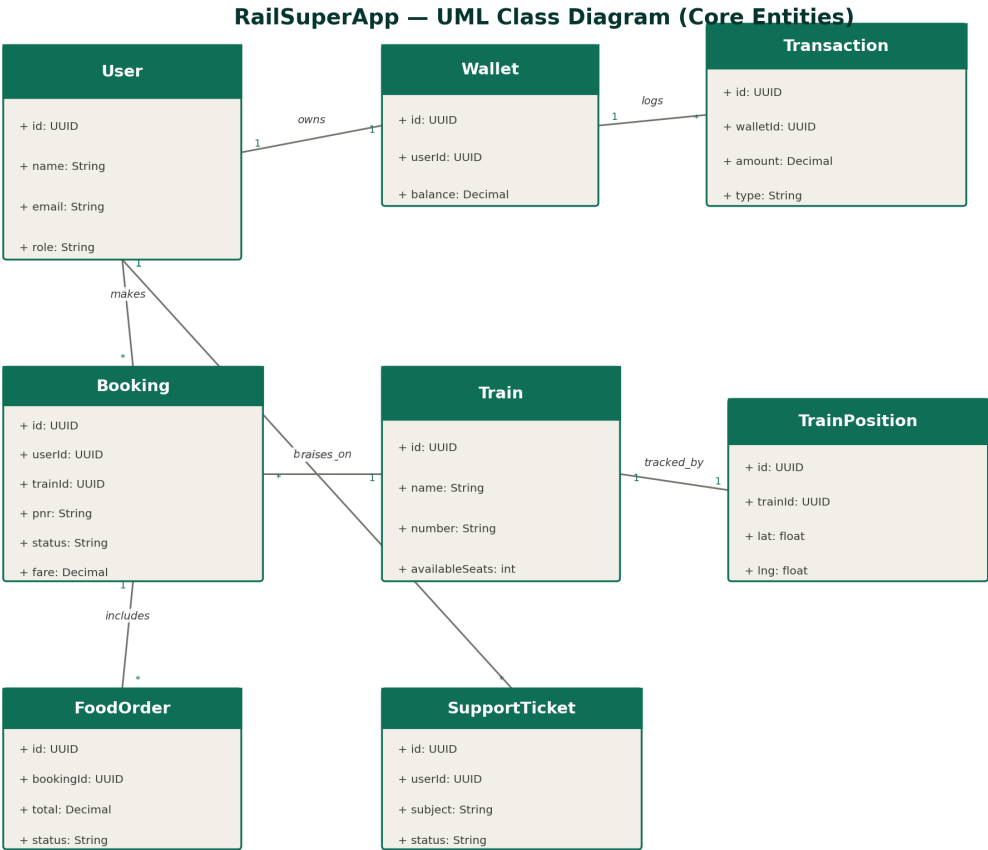


Figure: UML class diagram of RailSuperApp's core domain entities and relationships.

Sample JSON Data

POST /api/auth/login — Request

```
{
  "email": "leo@example.com",
  "password": "SecurePass123"
}
```

POST /api/auth/login — Response

```
{
  "token": "eyJhbGciOiJIUzI1NiIs... ",
  "userId": "b3f1a2c4-1234-4a5b-9c8d",
  "name": "Leo"
}
```

GET /api/trains/search — Response

```
[
  { "id": "t1", "name": "Rajdhani Express", "number": "12951",
    "fromStation": "New Delhi", "toStation": "Mumbai Central",
    "frequency": "DAILY", "availableSeats": 319, "baseFare": 850.00 }
]
```

```
]
```

POST /api/bookings — Response

```
{
  "id": "bk_9001", "pnr": "PNRA2AC64DE", "status": "CONFIRMED",
  "fare": 850.00, "seatNumbers": ["S1-24"]
}
```

GET /api/wallet/transactions — Response

```
[
  { "id": "txn_2201", "type": "CREDIT", "amount": 500.00, "description": "Wallet top-up" },
  { "id": "txn_2200", "type": "DEBIT", "amount": 850.00, "description": "Ticket booked - PNRA2AC64DE" }
]
```

POST /api/chatbot/message — Response

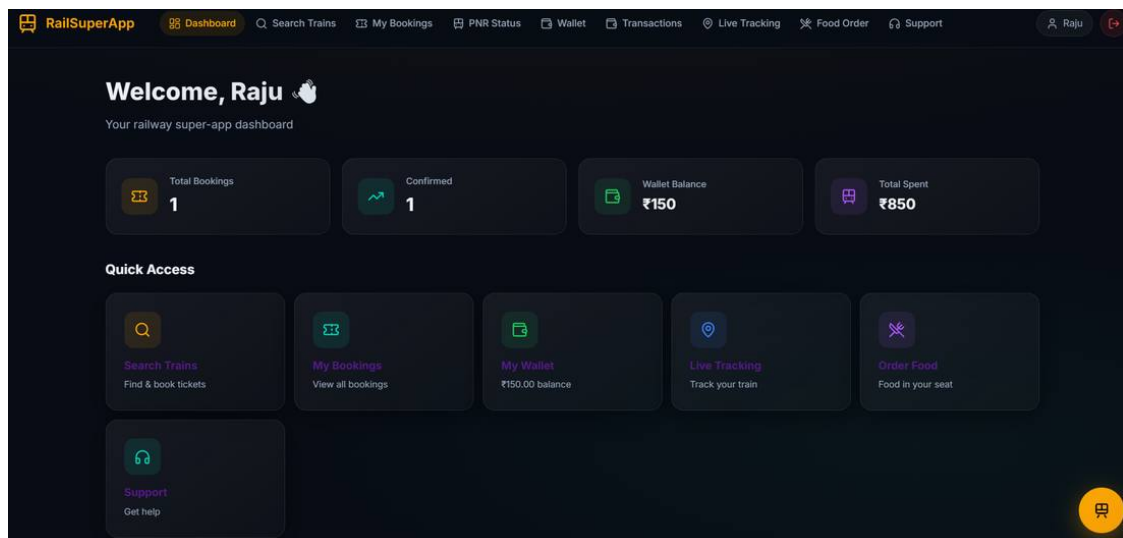
```
{
  "reply": "Train 12951 is currently near Alwar, running 12 minutes late.",
  "intentType": "LIVE_TRACKING"
}
```

Default User Details

Field	Value
Name	Leo / Raju (demo accounts)
Email	leo@railsuperapp.com
Password	Leo@1234
Role	PASSENGER
Initial Wallet Balance	Rs. 1000.00

Note: Placeholder demo credentials for local testing only.

Demo Pictures



Demo Picture 1: Dashboard view

RailSuperApp

DashboardSearch TrainsMy BookingsPNR StatusWalletTransactionsLive TrackingFood OrderSupport

Raju

Search Trains

Find available trains between stations

FROM STATION
New Delhi (NDLS)

TO STATION
Mumbai Central (BCT)

TRAVEL DATE
23-07-2026

NUMBER OF SEATS
1

Search Trains

2 trains found

Rajdhani Express
#12951 • RAJDHANI
Runs Daily

16:25
NDLS

08:15
BCT

From
₹850
319 seats left

Book

Rajdhani Express (Evening)
#12951-2 • RAJDHANI
Runs Daily

16:25
NDLS

08:15
BCT

From
₹1850
500 seats left

Book

Demo Picture 2: Search Trains results

My Bookings

View and manage all your train tickets

Rajdhani Express (12951)
PNR: PNRA2AC64DE
CONFIRMED

Journey
New Delhi → Mumbai Central
21 Jul 2026

Details
SLEEPER • 1 Seat(s)
Fare: ₹850.00

Passengers
raju (25)

Order FoodCancel Ticket

Demo Picture 3: My Bookings

PNR Status

Check the current status of your train ticket

PNRA2AC64DE

Check Status

PNR: PNRA2AC64DE
CONFIRMED

Rajdhani Express (12951)
New Delhi → Mumbai Central

JOURNEY DATE
Sun, 19 Jul, 2026

TRAVEL CLASS
SLEEPER

TOTAL FARE
₹850.00

PASSENGERS
1

Passenger Details

Passenger 1
raju
25 yrs, MALE

Demo Picture 4: PNR Status

Transactions

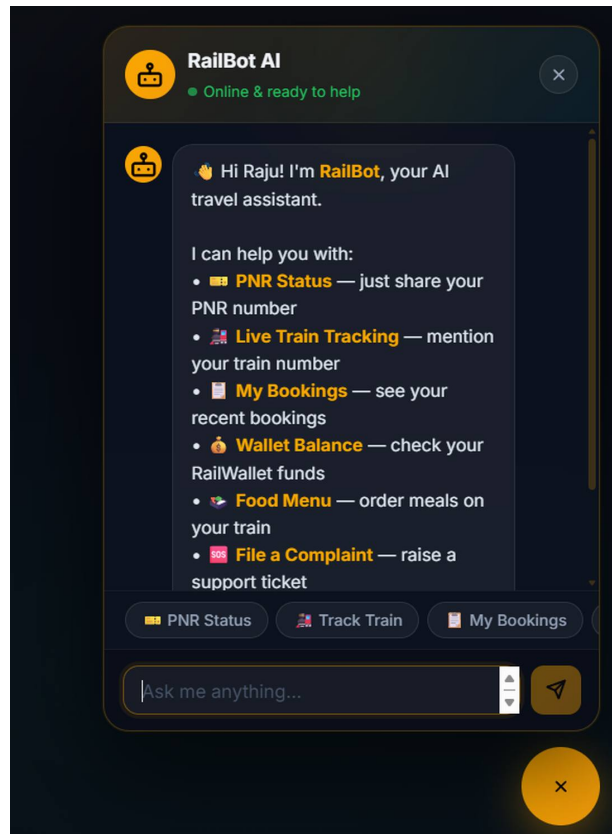
A detailed ledger of your wallet activities

Current Balance ₹650.00	Total Spent -₹850.00	Total Credited +₹500.00
----------------------------	-------------------------	----------------------------

Transaction Ledger

 Wallet top-up 21 Jul 2026, 2:37 pm	+₹500.00
 Ticket booked · PNR: PNRA2AC64DE · Train 12951 21 Jul 2026, 1:00 pm	-₹850.00

Demo Picture 5: Transactions ledger



Demo Picture 6: RailBot AI chatbot